Public Disclosure Authorized

Public Disclosure Authorized

Public Disclosure

Concept Environmental and Social Review Summary

Concept Stage

(ESRS Concept Stage)

Date Prepared/Updated: 06/18/2021 | Report No: ESRSC02037

BASIC INFORMATION

A. Basic Project Data

Country	Region	Project ID	Parent Project ID (if any)
Indonesia	EAST ASIA AND PACIFIC	P174350	
Project Name	Indonesia Sustainable Least-cost Electrification-1		
Practice Area (Lead)	Financing Instrument	Estimated Appraisal Date	Estimated Board Date
Energy & Extractives	Investment Project Financing	10/1/2021	3/31/2022
Borrower(s)	Implementing Agency(ies)		
PT PLN (Persero)	PT PLN Persero (PLN)		

Proposed Development Objective

The development objective is to increase access to least-cost electrification and sustainable electricity in Eastern Indonesia

Financing (in USD Million)	Amount
Total Project Cost	720.00

B. Is the project being prepared in a Situation of Urgent Need of Assistance or Capacity Constraints, as per Bank IPF Policy, para. 12?

No

C. Summary Description of Proposed Project [including overview of Country, Sectoral & Institutional Contexts and Relationship to CPF]

In 2019, PLN (with Bank support) developed the Indonesia Sustainable Least-cost Electrification project (ISLE Phase 1) Framework aims to reduce electricity generation cost in Eastern Islands with grid upgrades, improved planning, improved grid reliability, and mobilizing private investments. The ISLE Program consist of three phases: Framework Development, Preparation of Investments, and Framework Implementation in selected islands. The Phase 1 encompasses Bank executed upstream technical assistance and developed the framework & identified 21 potential investments across 10 islands. The start of Phase 2 to be financed under RETF (ISLE TA, P169259) has been delayed, and once started, the ISLE TA will finance feasibility studies (FS) of the identified investments and support further preparation if found to be feasible. As part of the FS, preliminary environmental and social (E&S) assessment will be conducted to inform the FS process. E&S instruments including Environmental and Social Management Plan (ESMP),

Public Disclosure

Labor Management Plan (LMP), Resettlement Action Plan (RAP), Stakeholder Engagement Plan (SEP), and Indigenous Peoples Plan (IPP) will be prepared according to the TOR cleared by the Bank for respective investments found to be feasible upon FS completion. The FS are expected to be finalized in mid 2022 and the E&S instruments by end 2022. The ISLE TA is expected to achieve grant signature in March 2021.

The Project, Phase 3 of the ISLE program, will not only implement the identified investments prepared under the ISLE TA, but will also prepare and implement other investments that had not been identified under the Phase 1 and scale-up the ISLE Framework to other islands. The Project components are:

a. Component 1 will support construction and operation of 86.5MWp PLN-owned solar PV in Alor, Rote, Buru, Morotai, Seram and Tual/Kei Kecil to be prepared under ISLE TA, and provide technical assistance (TA) to selected Independent Power Producers (IPPs) in Timor, Sumbawa, Ternate, and Flores, with focus on providing transaction support and advice to attact private participation in solar projects development.

b. Component 2 will finance construction of two 80km 150kV backbone transmission lines in Flores and Timor, provision of variable renewable energy (VRE) integration and reliability equipment (SCADA system, step-up transformers, 20 kV doubling) and 180 MWh battery storage across 10 islands to be prepared under ISLE TA.

c. Component 3 will finance deployment of solar home system (SHS) in 10 islands, scale-up the SHS business to other 20-30 Eastern Islands, and pilot a revolving fund for 100,000 poor households to cover grid connection fees.

d. Component 4 will finance the scaling-up of the ISLE Framework to other 20-30 small grid islands including support on planning, FS, and E&S instruments; capacity building; and project implementation.

Public Disclosure

D. Environmental and Social Overview

D.1. Detailed project location(s) and salient physical characteristics relevant to the E&S assessment [geographic, environmental, social]

The Government of Indonesia is targeting universal access to electricity by 2024. About a third of the 6 million people without electricity are in Eastern Islands. The lowest electrification rates are in Nusa Tenggara Timur (NTT) with 86%, South Maluku with 91% and Central Kalimantan and Central Papua with 94.5%. The 10 islands where 21 potential investments (i.e. PLN solar PV, transmission lines, VRE integration & reliability equipment and battery storage) had been identified under ISLE Phase 1 are located in three Eastern Regions, namely, Maluku (Morotai, Tual/Kei Kecil, Ternate, Seram, Buru), Nusa Tenggara Barat (Sumbawa) and NTT (Alor, Rote, Flores, and Timor Barat). The FS to be conducted under the ISLE TA will assess the feasibility of 21 potential investments for the final selection of investments under the Project. The preliminary ES assessment is expected to be completed in mid-2022, during the implementation of this Project. The 10 islands where 21 potential investments had been pre-identified were selected by PLN based on the following criteria: (i) PLN grid size (5-250 MW); (ii) average generation cost; and (iii) existing electrification rate. Preliminary ES assessment of investments that had been identified so far will be conducted as part of FS to be funded under the ISLE TA. The Project will also finance the scale-up of SHS business and the ISLE Framework in other 20-30 Eastern Islands, to be selected during implementation using similar criteria.

Overall, the Project E&S risks are moderate to substantial, especially as some investments under the scale-up of the ISLE Framework are yet to be identified. The 10 islands where 21 potential investments had been pre-identified, as well as additional 20-30 Eastern Islands where more investments will be identified, are home to rich biodiversity, diverse natural habitats and ecosystems, and cultural diversity. Maluku comprises 8 major islands and at least 10 islands groups. Nusa Tenggara Barat (NTB) has two main islands beside the smaller islands, while the NTT has many islands. Maluku and NTB are lagging behind other provinces due to their dry climate, soil conditions, and the high cost of transportation and infrastructure development due to geographical condition. Most people live in rural communities and derive livelihoods from natural resources e.g. farming, fishing, and cattle rearing and depend on forest products including non-timer forest products. They are highly vulnerable to natural disasters such as drought that could affect crop yields and livestock. The regions also own rich nature and cultural diversity, resources and customs. There are Indigenous Peoples (IPs) with a variety of traditions and customs expressed in distinct languages, dialects, local wisdoms, and arts. They live in remote rural locations with limited access to infrastructure. In Maluku, majority of local population are Tobelo and Galela sub-ethnic groups who migrated from North Halmahera following past natural disasters. There are IP communities in Buru and Seram including in Sub-district (Kecamatan/Kec.) Waeapo (Wailua, Mual, Bihuku, etc.) and Namlea (Rana and Wahidi) in Buru, and Kec. Kairatu (Waemali, Alune, Rambatu) in Seram. There is Samawa in Kec. Batulanteh, Sumbawa District of NTB. Lastly, the NTT remains one of the least developed provinces in Indonesia and has IPs community such as in Alor (Maran Abui, Klon) and in Flores Timur. The protected or endangered species in these regions are mostly marine species, birds, and terrestrial species such as Komodo Dragon of which their habitats are mostly quite remote from the targeted area for this Project.

Investments in Eastern Islands will be identified during implementation, but they will be in or adjacent to the Maluku, NTB, and NTT regions, and will have similar characteristics with those that had been identified in terms of biodiversity and natural habitats, and ethnic, cultural and livelihoods diversity of local population.

D. 2. Borrower's Institutional Capacity

PLN has implemented several Bank funded projects in the recent past. Besides the ISLE Phase I under which the ISLE Framework was developed and 21 potential investments identified, the PLN implemented Indonesia Power Transmission Development (IPTD) Project Phase I (P117323) and II (P123994), in which the PLN conducted FS Stage ESIA, CIA, and LARAP for Pokko Hydropower Plant, in addition to the extension and rehabilitation of 150 kV and 70 kV substations and new construction of 150 kV substations. The PLN also implemented Pumped Storage TA project (P112158) to prepare the ESIA for Upper Cisokan Pumped Storage (UCPS) and for Matenggeng Pumped Storage (MPS), however, the former was dropped due to low quality and significant delay. The preparation of the ESIA of the MPS, expected to be completed in December, 2021, has been delayed. Under the Development of Pumped Storage Hydropower Jawa Bali System project (P172256), the PLN would complete the ESIA and other instruments for the UPCS and finance its construction, and prepare Detailed Design and E&S instruments of Matenggeng Pumped Storage and Pokko Hydropower plant. Lastly, the implementation of the ISLE TA (P169259) is expected to start soon under which the Feasibility Study (FS) of investments to be financed under the Project would be conducted, and a preliminary E&S assessment (as part of the FS) and relevant E&S instruments (ESMP including LMP, RAP, and IPP) be develoed in line with the Bank ESSs. Efforts to strengthen the overall PLN's Environmental and Social Management System will also be made as part of ISLE TA activities.

Enhanced assistance and support by the World Bank has been required in order for PLN to demonstrate a satisfactory E&S performance on many of these operations. Furthermore, the ongoing work that PLN has been conducting in collaboration with ADB based on the Country Systems has highlighted PLN's needs for capacity development.
PLN continues to build its internal E&S capacity in terms of staffing and their qualifications. PLN staff in Head Quarters (HQ), as well as 19 PLN staff from regional offices, have taken the ESF roll-out training. At this moment, every Regional Project Office (Unit Induk Pembangunan-UIP) has dedicated E&S staff at regional offices and last year PLN initiated the establishment of an Health, Safety, Security and Environment (HSSE) academy. Furthermore, PLN has allocated for this project a project coordinator, an environment specialist, a social safeguards specialist, a procurement specialist and a financial management specialist (FM). Additionally, five main divisions of the PLN are

Public Disclosure

following the project, including PLN Planning Division – corporate and system planning, PLN Wilayah – regional entities, PLN Procurement Division, PLN Safeguards Division, and PLN Renewables. The overall Environmental and Social Management System at corporate level still needs strengthening in term of allocation of role and responsibility, internal and external communication and capacity building. Strong coordination between PLN Head Office and PLN Wilayah will be needed to ensure the success of the project. The capacity support will be located within the planning teams.

II. SCREENING OF POTENTIAL ENVIRONMENTAL AND SOCIAL (ES) RISKS AND IMPACTS

A. Environmental and Social Risk Classification (ESRC) Substantial

Environmental Risk Rating Substantial

The Project will create positive environmental benefits by reducing GHG emission and air/water pollution from diesel use in remote areas. Environmental risk is substantial as the project may result in medium adverse environmental risks to sensitive receptors. The environmental risks of PLN solar PV/battery storage, transmission lines, VRE integration & reliability equipment are not likely to be significant, irreversible & unprecedented since they are not complex/large scale and site specific. Potential construction impacts are dust & reduced local air quality, soil & water resources contamination and depletion, noise pollution, habitat loss due to footprints of solar PV/battery storage and natural habitats fragmentation and/or alteration due to transmission lines routing. These impacts can be avoided, minimized/mitigated by alternative site location assessment, adoption of state-of-art technologies, and good engineering design. There is low probability of adverse effects on community and occupational health and safety from transport of equipment and installation, e.g. electrocution risk when household connections are done by unqualified personnel. Potential operation impacts are use of water resources, hazardous waste generation, fire & explosion risk from faulty wiring and battery storage systems, overhead risks to birds and bats and electric magnetic fields impacts from transmission lines. Solar PV decommissioning may result in soil & water contamination due to damaged/improperly disposed of solar panels. The impacts are low to moderate significant, reversible and localized in nature, and can easily be mitigated by proper handling of used solar panels in collaboration with primary supplier. Preliminary E&S assessment and ESMP for these investments will be prepared under ISLE TA to ensure measures are in place. This Project will include components not covered in ISLE TA. The TA to IPP solar PV will not involve civil works and the site locations are not yet known, but potential downstream impacts from future investments are low to moderate significant similar with the PLN solar PV. E&S consideration will be part of location selection criteria to avoid adverse impacts and bidding documents will incorporate requirements consistent with ESF. SHS deployment & scale-up of SHS business will not likely result significant impacts on natural habitat since the installation is in household level. But SHS operation will require regular spare parts replacement which some are categorized as hazardous waste. The impacts will be predictable, site specific, and can be mitigated with proper handling of used spare parts in collaboration with SHS supplier. Locations of scaling-up of ISLE framework are not yet identified but expected to be adjacent to the 10 islands, thus will likely to have similar receptors' sensitivity. The potential risks are expected to be medium in magnitude and in spatial extent, and mostly temporary, predictable, and reversible. Cumulative impact assessment will not be required since the investments will be distributed to 10 different islands with no past and future project located cumulatively at single location. Where multiple types of investments situated in the same island, they will potentially be impacting different Valued Environmental Components (VECs). However, this will be confirmed during appraisal. The principles and guidelines to assess E&S risks and impacts of

Public Disclosure

each project component will be set in the ESMF, including provision to carry out appropriate environmental assessment commensurate to the risk. The ESMF will contain generic measures to avoid, reduce, mitigate, and/or offset adverse risks & impacts and guide impact management & monitoring plan preparation and reporting requirements as per national regulations, ESF, and WBG EHS guidelines. PLN E&S capacity building requirements will be further assessed during project preparation and capacity building plan will be included in the ESMF.

Social Risk Rating Substantial

The social risk is substantial due to significant land acquisition; IP community presence; and limited implementing agency capacity. Individual investments will require less complex, less intensive resettlements, and impacts on high value or sensitive areas are readily avoidable. Land acquisition will be small for power poles (Sub-Component 2.1), minor expansion of PLN's existing facilities to accommodate expanded battery capacity (Sub-component 2.2) and small capacity solar systems (Sub-Component 3.1), or substantial for hybrid system installations (Sub-Component 1.1) and large-scale solar projects financed by the private sector (Sub-Component 1.2). A preliminary E&S assessment will be conducted under the ISLE TA to identify, and avoid or minimize, E&S impacts. Since the exact project footprint is not known before appraisal, an RPF will be prepared to describe a Voluntary Land Donation (VLD) and willing buyer-willing seller protocols, and processes and procedures for involuntary land acquisition. Hybrid system installations will use government/PLN-owned land to the extent possible. If government-owned land is not available, the project may establish a lease agreement with the right to build (Hak Guna Bangunan – HGB) with land owners on a willing-buyer-willing seller basis and/ or VLD, following ESS5. Existing tenure rights and use of lands will be assessed, and lands where multiple claims to tenure rights or legacy issues exist will be excluded. The project will avoid construction of hybrid systems on communal lands, however, on an exceptional basis, a lease agreement may be established where community land titles are firmly established, broad community support for such lease arrangements are confirmed based on adequate consultation processes, and benefit sharing mechanisms agreeable to community members are established. No VLD or willing-buyer willing-seller land transaction will be allowed for communal lands. The RPF and the IPPF will provide details about the screening and eligibility determination protocol. For large-scale solar projects, involuntary land take may apply and a RAP will be prepared. No VLD is allowed. Like for hybrid systems, lands with multiple claims to tenure rights or legacy issues will be excluded, and the construction will be allowed on communal lands only where the principles mentioned above are followed, per provisions in the RPF and the IPPF.

Public Disclosure

B. Environment and Social Standards (ESSs) that Apply to the Activities Being Considered

B.1. General Assessment

ESS1 Assessment and Management of Environmental and Social Risks and Impacts

Overview of the relevance of the Standard for the Project:

ESS 1 is relevant. The Project will generate positive environmental benefits from reducing GHG emission and air/water pollution due to reduced use of fossil fuel (diesel) at remote areas through investments in renewable energy. However, the project will also potentially result in adverse environmental impacts from implementation of the overall Project components. The poor design risk and siting choices of the physical investments as well as from construction and operation activities present low to moderate significance environmental, social, health and safety risks for the project workers. The construction of physical investments will potentially create negative impacts on local air quality, soil and water resources, noise pollution and on natural habitats from medium to large footprints of

the solar PV/battery storage facilities or from the medium voltage transmission line construction. The construction of transmission line may also cause habitat alteration and habitat fragmentation. There is low probability of serious adverse effects to human health, mainly associated with community and occupational health and safety during installation, e.g. electrocution risk particularly due to household connections by unqualified personnel. Transportation of equipment into remote areas may require road safety and community awareness measures, involving labor influx. Interactions between contracted workers and local communities are likely to be infrequent and small scale, and manageable by codes of conduct. However, construction activities, including risks associated with labor influx, may increase the risks for Covid-19 infection to local community. A strict Covid19 prevention protocols will be in place per the government's Covid-19 protocols and Bank's Covid-19 health measures at construction sites. The operation of the physical investments may involve use of water resources for cooling or cleaning processes in solar plants & generation of hazardous waste from used battery & panel replacement, and potentially result fire & explosion from faulty wiring and battery storage systems and overhead risks to birds and bats and impacts from electric magnetic fields from operation of transmission lines. Additionally, the project may potentially impact ecosystem services mainly related to the footprint of solar PV, including on provisioning services due to lost of habitat and regulating services related to interference with rainfall and drainage. These potential impacts are low to moderate in intensity/significance and are reversible and localized in nature and therefore, can properly be mitigated. A financing mechanism will also be set up to provide poor and vulnerable people in villages that have existing access to electricity supply an financial support for the payment of initial connection fees. Only minor refurbishments such as wiring within houses or installation of meters are expected, with negligible environmental impacts. Implementation arrangement and decision procedures including eligibility criteria and refurbishment protocols will be described in the ESMF, and further spelled out in the Project Operations Manual (POM).
There is a risk that vulnerable people in the project areas cannot access project benefits, which will be managed through financial support in the payment of initial connection fees. The preliminary E&S assessment will identify vulnerable groups in the area of influence of respective investments, assess their ability to access revolving funds, affordability and availability of tariffs, community capacity to help poor households obtain and maintain access in an inclusive and participatory manner, the likelihood of residual exclusion (poorest households not able to join at all), lack of financial awareness and risk of indebtedness, and so on. Key principles to ensure inclusion of vulnerable and excluded groups and minimize elite capture, as well as implementation arrangements and decision procedures, will be included in the ESMF, to be further spelled out in the POM. Relevant international experience such as Power to the Poor program in Laos and Myanmar will be assessed. Livelihood impact is possible as most people in project areas depend on natural resources for their livelihood which the project may restrict, which will be assessed during project preparation and implementation. Mitigation measures will be included in the ESMF.
The E&S risk management for Sub-Component 1.1 and Component 2 is mainstreamed in the design and siting of investments as E&S assessment is integrated in the feasibility studies (FS) to be financed by ISLE TA. The start of ISLE TA has been delayed due to procurement issues. Once started, ISLE TA will finance FS and preliminary E&S assessment of 21 investments in 10 islands that had been identified under the ISLE Phase 1. The preliminary E&S assessment will not substitute necessary environmental assessments, such as AMDAL (or full ESIA) or UKL-UPL (partial environmental assessment), required under national laws to obtain relevant environmental permits. Following the completion of the FS, E&S instruments including Environmental and Social Management Plan (ESMP), Labor Management Procedure (LMP) Resettlement Plan (RAP), Stakeholder Engagement Plan (SEP) and Indigenous Peoples Plan (IPP) will be prepared according to the TOR that has already been cleared by the Bank for respective investments if found to be feasible under ISLE TA. The ESCP will include a requirement that the instruments mentioned above will be prepared according to the TOR cleared by the Bank. The ESMP will contain specific

Public Disclosure

management information to the relevant ESSs at specific locations and provide a plan and detailed procedure to be followed by PLN to manage and mitigate potential risks and impacts during physical investment, to ensure that measures are in place for pre-construction, construction and operation phase. The preliminary E&S assessment is expected to be completed in mid-2022. The FS are expected to be finalized in mid 2022 and E&S instruments are expected to be finalized end 2022. The ISLE TA is expected to achieve grant signature in July 2021.
Since ISLE TA will unlikely complete before appraisal of this Project, an ESMF will be prepared and disclosed by PLN before appraisal which will set out the principles and guidelines to assess the E&S risks and impacts of overall Project's activities. The ESMF will contain generic measures to avoid, reduce, mitigate, and/or offset adverse risks and impacts, and information on the responsible party/unit for addressing project risks and impacts and its capacity to manage risks and impacts. The ESMF will include list of typical E&S issues and risks for each project components, and guide the preparation of E&S screening, impacts assessment commensurate to the risk, and impact management & monitoring plan and reporting requirements for each project components in accordance with national laws and regulations, ESF, and WBG EHS guidelines. Environment Code of Practices (ECOP) will be annexed to the ESMF to guide planning & implementation of mitigation measures for civil works and installation of equipment, and handling of hazardous material and disposal of hazardous waste. The ESMF will also provide TOR for preparation of E&S assessment and the typical ESMP for scaling-up the ISLE Framework. The ESMF will also include a protocol to avoid and address SEA/SH incidents, including application of a code of conduct. The Project will ensure that (i) PLN will hire and retain dedicated E&S specialists to ensure the Project compliance with the ESCP including adequate implementation of E&S instruments and monitoring and reporting of E&S performance; (ii) PLN will ensure inclusion of E&S instruments into tender documents; and (iii) PLN E&S Specialists will provide E&S trainings to all PLN's staff involved in Project implementation, especially PLN Regional, and other relevant entities as necessary.

Areas where "Use of Borrower Framework" is being considered:

Not applicable.

Public Disclosure

ESS10 Stakeholder Engagement and Information Disclosure

Since the exact list of investments will not be determined before Appraisal, a Stakeholder Engagement Framework (SEF) will be prepared during preparation to guide the development of the Stakeholder Engagement Plan (SEP) at the investment level. Stakeholders including local population will be engagaged meaningfully as part fo the FS process which will start during the preparation of this Project. With the Covid-19 condition, the SEF will include a guidance on public consultation/stakeholder engagement in line with Government of Indonesia on Covid19 protocols as well as Bank's Covid-19 health measures at construction sites in order to minimize the risk of COVID-19 infection among contracted workers. Where there are constraints, stakeholder engagement activities will need to be adjusted to ensure an effective and meaningful consultations to meet project and stakeholder needs with minimum risks to expose them to Covid-19. The investment-specific SEP will be prepared and implemented per the SEF. The SEP will help PLN build and maintain over time a constructive relationship with their stakeholders, including local beneficiaries and affected communities. Key stakeholders, including vulnerable and excluded population, will be identified and consulted for respective investments after they are identified under the FS process, in line with the SEF. The SEP will ensure that beneficiaries and affected communities will be engaged, especially regarding electrification options, designs and locations, and the eligibility criteria and applicable rules of Grid Connection Fees Fund. It will also take into account the differential needs and participation requirements of men, women and those more likely to be excluded from participation due to their circumstances as the project location would be remote small islands. Processes and procedures to ensure meaningful engagement with IP groups will be described in the SEF

Public Disclosure

to guide the development of the investment-specific SEP, consistent with ESS7 and the Project IPPF. Relevant IPs representative organizations will be consulted during the preparation of the site-specific ESMP and IPP. A stakeholder grievance mechanism (GRM), as part of SEP, will be formed for the project so that project stakeholders including those who believe are negatively affected by the project can raise their concerns and have them addressed per project E&S instruments, to be documented in the grievalce log to be prepared for the projct.

Draft ESMF and SEF will be disclosed publicly in the PLN website and regional offices of PLN at these three regions, including AMDAL (or full environmental social assessment or ESIA) if necessary needed for specific sub-projects that meet the requirements in the Kepmen 38/2019 or has potential significant E&S adverse impacts. Investment specific instruments such as the ESMP, LARAP, IPP and SEP, will be disclosed and consulted locally in a language understandable to local stakeholders including but not limited to project affected people. Availability of and access to Stakeholder Grievance Mechanisms will be announced widely to the public to allow stakeholders to lodge a grievance about or feedback to project activities. The compliance requirement with regard to investment specific SEPs and their implementation will be included in the ESCP.

B.2. Specific Risks and Impacts

A brief description of the potential environmental and social risks and impacts relevant to the Project.

ESS2 Labor and Working Conditions

ESS2 is relevant. it is expected that Direct Workers and Contracted Works are relevant to the project. The relevance of primary supply workers is under assessment. Community Workers are not expected to be relevant. Environmental and social screening activities prepared by the ISLE TA as well as carried out during project implementation as required by the ESMF, will consider risks and impacts of proposed investments on labor and working conditions, as well as draft provisions and labor procedures for later consideration in the design and tender process. During prparation, PLN's own labor management procedures including management of contractors will be assessed to identify gaps with ESS2, and a Labor Management Procedure (LMP) will be prepared based on the result before appraisal. The LMP will address issues including OHS, equal opportunity treatment, a grievance mechanism specifically for workers, contractor management, prohibitions on harmful child and forced labor, among other issues, which will be incorporate in the ESMF where relevant. The LMP will also include measures to minimize risks related to Covid-19 infection to Contracted Workers in line with Government of Indonesia on Covid19 protocols as well as Bank's ESF/Safeguards Interim Note: COVID-19 Considerations in Construction/Civil Works Projects.

ESS3 Resource Efficiency and Pollution Prevention and Management

ESS3 is relevant. The Project will promote renewable energy related investments which will contribute to GHG emissions reduction. However, the project will also potentially result in environmental pollution from potential improper handling of hazardous material and disposal of hazardous waste procured and used as part of project activities (such as used batteries handling, used SHS spare parts handling, and other hazardous waste materials). Additionally, from all types of investments, solar power generation and transmission lines are more likely to generate greater E&S risks and impacts. Typical main E&S issues for solar power generation includes the size of land required for construction of power generation components and facilities, impacts on human health and safety from construction activities of the physical investments, the use of water resources for cooling or cleaning processes in solar plants, and handling of hazardous waste from used battery and panel replacement. While for the transmission lines, the linear type of development may required less land, but it is susceptible to generate impacts from electric magnetic fields.

Preliminary E&S assessment and ESMP prepared as part of the ISLE TA shall assess the possible impacts and risks of the investments identified under Phase 1 (solar PV and battery storage, and transmission lines) on environmental pollution from procuring goods and materials (such as used batteries handling and other hazardous waste materials), impacts on human health and safety and the possible construction impacts to environment. Additionally, the ESMF will include guidelines to assess the environmental risks and impacts for each project components, in particular related to procurement, storage, handling, and safe disposal of hazardous material, especially from the operation of SHS and solar PV. The ESMF will also specify national regulations requirements related to storage, transport, and disposal of hazardous materials and hazardous waste and will emphasize collaboration with SHS and solar panel supplier for recycling and/or disposal of used SHS spare parts and solar panels. The ESMF will also include screening mechanism and analysis of the potential impacts and mitigation measures for resources efficiency and pollution prevention aspects of each project activities. The compliance requirement with regard to the ESMF implementation as well as preparation of investment-specific ESMP will be included in the ESCP.

Public Disclosure

ESS4 Community Health and Safety

ESS 4 is relevant. The Project will include physical investment that may cause potential impacts and risks on community health and safety, especially from construction activities, transportation of equipment into remote areas, interactions between contractors and remote local beneficiaries and affected communities, electromagnetic fields from transmission line (perceived or real), and possible impacts related to ecosystem services. Risks associated with labor influx are considered moderate as constructions are expected to be small to medium size. While the project plans to mostly employ local labor, with a small number of external workforce that will be outsourced for specialized skills, local absorption capacity is low. Using the SEA/SH risk screening tool, the SEA/SH risk rating of this project is assessed as low. Other community health and safety risks include risk of electrocution (particularly when household connections are done by unqualified workers and community members), fire and explosion from faulty wiring and battery storage systems. The construction activities, including risks associated with labor influx, may increase the risks for Covid-19 infection to local community. A strict Covid19 prevention protocols will need to be in place in line with Government of Indonesia on Covid19 protocols as well as Bank’s Covid-19 health measures at construction sites. Additionally, possible impacts related to ecosystem services will mainly related to the footprint of solar PV, including potential impacts to provisioning services due to lost of habitat and potential impacts to regulating services related to interference with rainfall and drainage. These potential impacts are moderate in intensity/significance and are reversible and localized in nature and therefore, can be properly mitigated. Preliminary E&S assessment will be conducted and ESMPs be developed under the ISLE TA for the investments identified under Phase 1 (solar PV and battery storage, and transmission lines) according to the TOR cleared by the Bank, which will unlikely be completed before Appraisal. Additionally, the ESMF will include guidelines to assess the community risks and impacts for each project components, in particular related to deployment of SHS as well as the downstream impacts from construction and operation of IPP-owned solar-PV and future investments from scaling-up of the ISLE Framework. The ESCP will clearly provide the PLN's responsibility to carry out E&S screening, impact assessment, and impact management and monitoring plan (ESMP and other E&S instruments) as per the ESMF in compliance with the ESF, and incorporate relevant E&S risk mitigation measures in respective contracts of investments and/or civil works contract to the extend possible. All ESMPs to be prepared under the Project will provide measures to avoid, reduce, and mitigate risks related to environmental health and safety (EHS) and Community Health and Safety (CHS) issues including road safety, labor influx, electrocution, fire and life safety, SEA/SH risks. The contractor will be required to develop their own codes of conducts in line with the investment specific ESMP to govern interactions between contracted workers and local communities. Commitment to comply with the ESMF for the overall project as well as with the ESMP for specific investments are included in the ESCP.

ESS5 Land Acquisition, Restrictions on Land Use and Involuntary Resettlement

ESS5 is relevant. Project components may lead to land taking for the proposed transmission line in Flores and Timor, solar PV projects with battery investments, and grid upgrades. The amount of land needed is expected to vary from limited for erecting power poles, grid upgrades (transformer, SCADA etc.) and SHS, to substantial for the hybrid system installation or stand-alone large-scale PV project. While the exact size of lands required for respective investments are not known, it is expected that medium size investments (in the range of 10 – 35MW installed capacity) would require about 10 – 35ha of lands while large-scale investments (about 100 MW installed capacity) would require about 120ha of lands. It is expected that one medium scale investment would be implemented in each of the six islands that have been identified, while two large scale investment would also be implemented in two selected islands. The installation of Hybrid system installations will use government/PLN-owned land to the extent possible to mininize impact on private lands.

The project will result in loss of lands or access to lands. Since the exact project footprint is not known before appraisal, a Resettlement Policy Framework (RPF) will be prepared during preparation that will describe a Voluntary Land Donation (VLD) Protocol and willing buyer-willing seller protocol, and processes and procedures to be followed for involuntary land acquisition.

Hybrid system installations (Sub-Component 1.1) will use government/PLN-owned land to the extent possible. However, if government-owned land is not available, the project may need to establish a lease agreement with the right to build (Hak Guna Bangunan – HGB) with land owners on a willing-buyer willing-seller basis (voluntary land transaction) and/ or VLD, following provisions of ESS5. Existing tenure rights and usage of lands where hybrid systems are to be built will be assessed prior to location detemrinations, and any lands where multiple claims to tenure rights or legacy issues are found to exist will be excluded. The project will avoid construction of hybrid systems on communal lands, however, on an exceptional basis, a lease agreement may be established where community land titles are firmely established, broad community support for such lease arrangements are confirmed based on adequate consultation nad participatory processes, and benefit sharing mechanisms agreeable to community members are established. No VLD or willing-buyer willing seller land transaction will be allowed for communal lands. The Resettlement Policy Framework (RPF) and the Indigeous Peoples Plannig Framework (IPPF) will provide details about the screening and eligibility determination protocol.

Public Disclosure

For large-scale solar projects (Sub-Component 1.2), involuntary land taking may apply and a RAP will be prepared. Private lands may be acquired on a willing-buyer willing-seller basis, but no VLD is allowed. Like for hybrid systems, lands where multiple claims to tenure rights or legacy issues are found to exist will be excluded, and the construction is allowed on communal lands only where community land titles are firmely established, broad community support for such lease arrangements are confirmed based on adequate consultation nad participatory processes, and benefit sharing mechanisms agreeable to community members are established. The RPF and the IPPF will provide details about the screening and eligibility determination protocol.

In applying VLD and willing buyer-willing seller schemes, a special care is required to ensure that the owners of the land must be able to refuse to sell or donate their lands if unwilling to sell or donate them without threat of compulsory acquisition, and fully informed about available choices and their implications. Furthermore, an exclusion list will be established in the RPF to prohibit physical displacement, including of IPs. Willing buyer-willing seller land transaction and voluntary land donation approaches will also be applied only when: (a) land requirement is small; (b) investment is a site-specific linear infrastructure construction where alternative siting is possible; and (c) land donors directly benefit from the investment. For willing buyer-willing seller schemes the owner should also be paid a fair price based on prevailing market values. The project will maintain a transparent record of all consultations and agreements reached.

Limited scale of land acquisition will occur under Component 2-1 to build towers for two, about 80km long, 150 kV transmission lines. Given the linear nature of the project, a large-scale physical displacement is unlikely. The exact size of land to be acquired for each tower as well as their locations will be known during implementation. Land use restrictions will likely be introduced below electricity cables that prohibit the use of lands within the safety zone for residential purposes, or planting of tall trees that may create safety hazards. Construction of towers on communal lands or where multiple claims exist will be avoided wherever possible. Component 2-2 will mostly be implemented within the existing, fenced premises of the PLN, however, minor land acquisition cannot be ruled out on an exceptional basis. The potential areas where the Component 2 may be implemented will be assessed during preparation and areas of high land acquisition risks will be avoided.

Public Disclosure

ESS6 Biodiversity Conservation and Sustainable Management of Living Natural Resources

ESS6 is relevant. The potential indirect impacts are related to characteristics of Project locations that are home to rich biodiversity with diverse natural habitats and ecosystems. The Project locations are home to a number of protected or endangered species, mostly related to specific marine species, birds, and also the endangered terrestrial species such Komodo Dragon of which their habitats are mostly quite remote from the targeted area for this Project. The potential impacts and risks are disturbance to terrestrial and aquatic biodiversity and ecosystem services from land clearing for construction of power generation components and facilities as well as operation of the solar PV and transmission lines. Additionally, the linear development of the transmission line may result habitat alteration and habitat fragmentation. The E&S screening, impact assessment, and impact management and monitoring plan prepared during Project implementation as well as as part of the ISLE TA and prior to any physical activities will assess the possible impacts to modified, natural, and critical habitats of the physical investments as well as from downstream investments. The site-specific ESMP will contain provisions for biodiversity assessment, management and conservation for each investment. Additionally, the ESMF will include guidelines on screening mechanism of natural habitats and biodiversity aspects. The ESMF will also include exclusion list to screen out activities that will cause significant conversion and degradation to modify, natural and critical habitats. The ESMF for the overall project as well as ESMP for each of the project components are included in the ESCP.

ESS7 Indigenous Peoples/Sub-Saharan African Historically Underserved Traditional Local Communities

ESS7 is relevant. The target locations of the project are in the Eastern Islands of Indonesia particularly in three regions: Maluku, West Nusa Tenggara – NTB, and East Nusa Tenggara - NTT, where the presence of IP groups who meet the criteria of the ESS7 is likely. In District of Buru and Seram have IP communities such as in Sub-district (Kecamatan/Kec.) of Waeapo (Wailua, Mual, Bihuku, etc.) and Namlea (Rana and Wahidi) in Buru, and Kec. of Kairatu (Waemali, Alune, Rambatu) in Seram. There is Samawa in Kec. Batulanteh, Sumbawa District. NTT has IP communities such as in Alor (Maran Abui, Klon), and in Flores Timur.
The Project sites will be in Eastern Indonesia and the initial assessment indicates IP presence although the exact list of IP communities who may be present where the Project will be implemented cannot be known before appraisal. An Indigenous Peoples Plannig Framework (IPPF) will be prepared that will provide processes and procedures for IP screening and consultation processes, and the development of the Indigenous Peoples Plan (IPP) if the project impact on their lands. Local communities that meet the criteria of IP community under the ESS7 will be recognized as IP communities and provision of the ESS7 will apply. The project will avoid construction of hybrid systems on communal lands, including the lands where IP communities are present. However, on an exceptional basis, a lease agreement may be established where community land titles are firmly established, broad community support for such lease arrangements are confirmed based on adequate consultation and participatory processes, and benefit sharing mechanisms agreeable to community members are established. No VLD or willing-buyer willing seller land transaction will be allowed for communal lands including lands where IP communities are present. The Resettlement Policy Framework (RPF) and the Indigenous Peoples Planning Framework (IPPF) will provide details about the screening and eligibility determination protocol. Where IP groups lose land or access to natural resources that they have traditionally occupied or had access as a resulf of the project, an Indigenous Peoples Plan (IPP) will be prepared including a Social Assessment (SA), and Free, Prior and Informed Concent (FPIC) will be sought with affected IP groups. Relocation of IP communities and signifiant impacts on their cultural heritage that is material to their ethnic, cultural, ceremonial or spiritual identity are unliekly and will be prohibited in the IPPF.

ESS8 Cultural Heritage

ESS8 is relevant. The project activities are likely to take place in areas with Indigenous Peoples and they have cultural significance to local communities, particularly natural features with tangible and intangible heritage significance including spirit sites associated with streams, caves, and grottos, etc. As the exact project locations are still unknown, to address unknown archaeological or historical remains and objects, including graveyards and/or individual graves, Chance Find Procedures can be assessed and included in the ESMF. The ESMF will include identification of cultural heritage and assessment of tangible and intangible significance in consultation with affected stakeholders, and deployment of a chance find procedure.

Public Disclosure

Public Disclosure

ESS9 Financial Intermediaries

ESS9 is not relevant. No FI involvement is envisaged in the project.

C. Legal Operational Policies that Apply

OP 7.50 Projects on International Waterways	No
OP 7.60 Projects in Disputed Areas	No

III. WORLD BANK ENVIRONMENTAL AND SOCIAL DUE DILIGENCE

A. Is a common approach being considered?

No

Financing Partners

N/A

B. Proposed Measures, Actions and Timing (Borrower's commitments)

Actions to be completed prior to Bank Board Approval:

Stakeholder Engagement Framework (SEF), Environmental and Social Management Framework (ESMF) including Resettlement Policy Framework (RPF), Labor Management Procedure (LMP), and Indigenous Peoples Policy Framework (IPPF), and Environmental and Social Commitment Plan (ESCP).

Possible issues to be addressed in the Borrower Environmental and Social Commitment Plan (ESCP):

Preliminary E&S Assessment, Environmental and Social Management Plan (ESMP), Resettlement Action Plan (RAP), Indigenous People Plan (IPP) and Stakeholder Management Plan (SEP) including Grievance Redress Mechanism (GRM).

C. Timing

Tentative target date for preparing the Appraisal Stage ESRS 30-Sep-2021

IV. CONTACT POINTS

World Bank

Contact:	Sabine Mathilde Isabelle Cornieti	Title:	Young Professional
Telephone No:	5220+34596 / 1-202-34596	Email:	scornieti@worldbank.org

Contact:	Puguh Imanto	Title:	Energy Specialist
Telephone No:	5781+3305 / 62-21-5299-3305	Email:	pimanto@worldbank.org

Borrower/Client/Recipient

Borrower: PT PLN (Persero)

Implementing Agency(ies)

Implementing Agency: PT PLN Persero (PLN)

V. FOR MORE INFORMATION CONTACT

The World Bank
1818 H Street, NW
Washington, D.C. 20433
Telephone: (202) 473-1000
Web: http://www.worldbank.org/projects

VI. APPROVAL

Task Team Leader(s):	Puguh Imanto, Sabine Mathilde Isabelle Cornieti
Practice Manager (ENR/Social)	Ann Jeannette Glauber Recommended on 11-Jun-2021 at 22:30:38 GMT-04:00
Safeguards Advisor ESSA	Ekaterina Romanova (SAESSA) Cleared on 18-Jun-2021 at 14:09:43 GMT-04:00

Public Disclosure